TSM 099 Polynomials Test

Practice 1 | Version 202610122020.19

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right.

QUESTIONS	ANSWERS
1. Factor completely over the integers, or write prime. $(a)\quad 2x^2 - 5x - 3$ $(b)\quad 2x^2 + x + 3$	1. (a) 10 pts (b) 10 pts
2. Multiply and simplify. $(2x - 3)(3x + 2)$	2. Polynomial 20 pts
3. Factor using the ac-method and grouping. Show ac, the split middle term, and the grouping. $12x^2 + 29x + 15$	3. ac; pair 4 pts Split middle term 4 pts Grouping 6 pts Factored form 6 pts
4. Factor completely. $(a)\quad x^2 + 2x - 8$ $(b)\quad x^2 - 25$	4. (a) 10 pts (b) 10 pts
5. Factor out the greatest common factor. $4x^4 - 2x^3 + 6x^2$	5. Factored form 20 pts

Instructor answer key: TSM 099 Polynomials Test

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Each column matches the vertical answer positions on its corresponding test page.

Concise grading reference; required work and checks are assessed on separate paper.

TEST PAGE 1

1. (a)

$(x-3)(2x+1)$

10 pts

(b)

Prime

10 pts

2. Polynomial

$6x^2-5x-6$

20 pts

3. ac; pair

$ac=180;\ 9,\ 20$

4 pts

Split middle term

$12x^2+9x+20x+15$

4 pts

Grouping

$3x(4x+3)+5(4x+3)$

6 pts

Factored form

$(3x+5)(4x+3)$

6 pts

4. (a)

$(x-2)(x+4)$

10 pts

(b)

$(x-5)(x+5)$

10 pts

5. Factored form

$2x^2(2x^2-x+3)$

20 pts